

MANAKSETU

From Verification to Trust

Smart India Hackathon (SIH)	Project Domain Digital Transformation of the Legal Metrology Verification and Certification Ecosystem
Core Proposition A unified digital ecosystem connecting instruments, people, verification workflows, certificates and administration.	One Platform Multiple Stakeholders • Complete Visibility • Better Compliance

1. Introduction

Legal Metrology exists to ensure that weighing and measuring instruments used in commercial activity remain accurate, verified and trustworthy. The verification and certification lifecycle, however, can involve multiple stakeholders, physical instruments, testing activities, applications, records and certificates. When these elements operate in fragmented workflows, users can face limited visibility, manual record handling, delays and difficulty in tracking validity and renewal.

ManakSetu addresses this gap by creating a **single connected digital ecosystem** for manufacturers, businesses and traders, citizens, Legal Metrology Officers (LMOs), Government Approved Test Centres (GATCs) and administrators. Instead of digitising only one form or one office workflow, it digitally connects the complete instrument lifecycle.

2. Problem Statement

Fragmentation	Stakeholder communication and records may exist across disconnected channels and workflows.
Limited lifecycle visibility	It can be difficult to trace an instrument from registration through verification, certification and renewal.
Status uncertainty	Businesses and citizens may have limited real-time visibility into verification progress and certificate validity.
Manual effort	Repeated data entry, physical records and scattered follow-ups can slow down processing.
Weak physical–digital linkage	A physical instrument may not have an easily accessible digital identity that exposes relevant verification information.

Need for a Unified Solution

The need is not simply for an online application form. The ecosystem requires a **shared digital backbone** that maintains a common instrument identity, routes requests to the right authority, records testing and verification, issues digital certificates, enables QR-based public verification, and gives administrators a central view of operations.

3. Proposed Solution — ManakSetu

ManakSetu is a **unified digital bridge** that connects every major stakeholder and every important stage of the Legal Metrology verification lifecycle. Each stakeholder gets a purpose-built portal, while the underlying platform maintains common data, role-based permissions, workflows and traceability.

The key innovation is the **ecosystem approach**: an instrument is not treated as a one-time application. It receives a connected digital identity that can follow it through registration, commercial use, verification, testing, certification, QR validation, renewal and compliance monitoring.

Complete Instrument Lifecycle

Manufacturer	Instrument Registration	Business / Trader Usage	Verification Request	LMO / GATC Processing	Testing & Verification	Digital Certificate	QR-Based Verification	Renewal & Compliance	Central Administration
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What ManakSetu connects	How the platform improves it
Stakeholders	Dedicated portals share a common platform, so actions and status changes can flow across roles.
Instrument identity	Structured instrument records create continuity from manufacturer registration to verification history.
Workflow	Verification requests move through defined processing, testing and certification stages with status visibility.
Certificate lifecycle	Digital certificate access, QR verification and renewal/compliance tracking reduce dependence on scattered records.
Administration	A central console provides monitoring, analytics and reporting for operational decision-making.

Why it is more than a simple portal

A conventional portal may digitise submission. ManakSetu goes further by **orchestrating the ecosystem**. The same digital record can support registration, assignment, testing, certification, public verification and renewal. This creates continuity, reduces information gaps and makes the complete lifecycle more traceable.

Core value proposition: One Platform. Multiple Stakeholders. Complete Visibility. Better Compliance.

4. Stakeholder Portals & Feature Set

ManakSetu uses role-specific portals so each stakeholder sees the information and actions relevant to them, while all portals remain connected to the same central data and workflow layer.

Stakeholder ecosystem

Customer & Citizen	Public-facing trust and transparency layer	QR scanning • relevant verification information • certificate visibility • consumer awareness • trust in verified instruments
Business & Traders	Operational compliance management	Instrument management • verification requests • application tracking • digital records • certificate access • history • renewal reminders
Manufacturer	Origin and registration of instruments	Instrument registration • structured specifications • digital traceability • linkage to the verification ecosystem
LMO	Official verification workflow	Request management • task assignment • instrument details • digital verification records • status updates • workflow management
GATC	Technical testing workflow	Testing activities • test records • digital reports • linkage between testing and certification
Administrator	Central governance and monitoring	User management • verification analytics • pending/completed activity • dashboards • reports • decision support

How information flows across portals

1	Manufacturer registers an instrument	Structured instrument record becomes available in the central ecosystem.
2	Business / trader uses and manages it	The instrument is associated with operational records and compliance needs.
3	Verification request is submitted	Request enters the workflow and becomes visible to authorised LMO/GATC users.
4	LMO / GATC processes the request	Testing, verification observations and status changes are digitally recorded.
5	Certificate is generated and linked	Certification becomes part of the instrument's digital history.
6	Citizen or inspector scans QR	The QR points to the relevant digital record and surfaces appropriate verification information.

Design principle: **role-based access control** ensures that stakeholders access only the information and functions relevant to their responsibilities, while the central system preserves end-to-end traceability.

5. Technology, Architecture & Digital Workflow

ManakSetu is designed as a layered digital platform. The architecture separates presentation, business logic and data storage, while APIs connect the stakeholder portals to shared services and workflows.

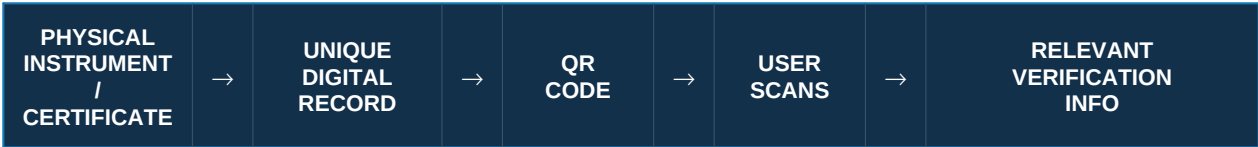
System architecture

FRONTEND LAYER User dashboards • forms • verification requests • status tracking • certificate access • QR interactions
↓ Secure API communication ↓
BACKEND / APPLICATION LAYER Authentication • APIs • business logic • workflow management • data processing • certificate handling • inter-portal communication
↓ Controlled data access ↓
DATABASE LAYER User records • instruments • manufacturers • businesses • verification requests • history • test results • certificates • QR records

Core technical capabilities

APIs	Standardised communication between portals and backend services; enables modular growth and integration.
Authentication & RBAC	Identity, role and permission checks ensure each stakeholder sees only authorised data and actions.
Workflow engine	Moves requests through registration, assignment, testing, verification, certification and renewal states.
Digital records	Maintains structured records for instruments, actions, tests, certificates and history.
QR identity bridge	Maps a physical instrument/certificate to a unique digital record that can be retrieved through a QR scan.

QR-Based Digital Verification — the physical-to-digital bridge



The QR mechanism is strategically important because it creates a simple bridge between the **physical world of instruments** and the **digital world of verification records**. It supports quick validation, transparency and easier access to the relevant verification information without requiring users to navigate the full administrative system.

6. Impact, Innovation, Scalability & Future Scope

ManakSetu is designed to create measurable operational value while strengthening public confidence in verified weighing and measuring instruments.

Expected impact

Transparency	Clearer status visibility and QR-accessible verification information improve confidence and reduce information gaps.
Compliance	Digital records, renewal reminders and lifecycle history make compliance easier to manage and follow.
Efficiency	Structured requests, digital workflows and central monitoring reduce avoidable manual coordination.
Traceability	Instrument, test, verification and certificate records remain connected across the lifecycle.
Citizen trust	A simple QR interaction can make relevant verification information more accessible at the point of use.
Administration	Dashboards and analytics support better prioritisation, monitoring and decision-making.

What makes ManakSetu innovative?

1. **Ecosystem, not form digitisation:** it connects six stakeholder experiences through a shared digital backbone.
2. **Lifecycle-centric identity:** the instrument remains the central object across registration, use, verification, certification and renewal.
3. **Physical–digital bridge:** QR technology turns the instrument/certificate into an accessible digital verification point.
4. **Role-aware governance:** RBAC aligns visibility and actions with stakeholder responsibilities.
5. **Operational intelligence:** the administrator console converts distributed activity into a central view for monitoring and decisions.

Scalability

The platform is designed to scale from **City** → **District** → **State** → **National**. A modular portal and API structure allows additional jurisdictions, users, instruments, centres and integrations to be added without redesigning the core concept. Centralised data models also support standardised reporting while permitting local operational workflows.

Future Scope

AI-based analytics and anomaly detection • predictive compliance alerts • automated SMS/email/app notifications • dedicated mobile application • integrations with government systems • advanced analytics dashboards • IoT-enabled monitoring where applicable • stronger digital certificate and identity services.

CONCLUSION

ManakSetu transforms Legal Metrology verification from a collection of disconnected activities into a **connected, transparent and traceable digital ecosystem**. By linking manufacturers, businesses, citizens, LMOs, GATCs, instruments, certificates and administrators, it builds continuity across the complete verification lifecycle.

MANAKSETU

"From Verification to Trust — Building a Transparent Digital Bridge for Legal Metrology."

One Platform. Multiple Stakeholders. Complete Visibility. Better Compliance.